Recently, alternative approaches have been proposed to explain LMs by intervening in the forward pass (Meng et al., 2022). When combined with token projection methods, this approach holds promise in providing insights into the "thinking" process of LMs (Ghandeharioun et al., 2024).

Our work ignores the additional scaling that is introduced by optimizers other than Stochastic Gradient Descent, such as Adam (Kingma, 2014). While the backward pass's VJPs remain unaffected when such optimizers are employed, they do alter the rank and weights of each gradient matrix, due to the additional scaling.

Our approach to explaining how knowledge is stored in LMs is grounded in single editing with a constant embedding. While our approach elucidates how models store various information, fine-tuning is typically conducted on multiple prompts and involves multiple steps (iterations). Additionally, training a model from scratch includes the training of its embeddings.

Our experimental approach to editing LMs with "forward pass shift" is presented as a case study rather than as a suggested alternative to existing methods. The results in Section 7, Appendix I might obfuscate "editing" and "output shifting", since only plotting the desired answers does not fully encapsulate the effect of the edit on similar prompts, which is a challenge faced by most editing benchmarks and datasets.

Our focus on the MLP layers excludes the attention layers. This decision is influenced by the growing consensus that MLPs are where LMs predominantly store information (Dai et al., 2022; Meng et al., 2022). We acknowledge the possibility that attention layers may also store information and that editing MLPs and attention simultaneously could have different effects on the model from those detailed in Section 5.2.

In the theoretical exploration, we omitted Layer Norms (Appendix B) and Dropouts to simplify the explanation, aligning with prior interpretability work that also overlooks these components due to their negligible impact compared to others (Elhage et al., 2021; Geva et al., 2023; Meng et al., 2023). While these components may influence gradients, our empirical results were conducted on full assumption-free LMs and show consistent patterns with the theoretical parts.

Lastly, this work was conducted on Decoder LMs with sequential architecture. It is important to note that other types of LMs might exhibit different behaviors in terms of their gradients. We chose this architecture as it is currently the leading architecture both in terms of downstream performance and in interpretability research (nostalgebraist, 2020; Brown et al., 2020; Touvron et al., 2023).

10 Ethics and Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here. However, future research could use the methods we developed to edit LMs. We hope such cases would be for developing better and safer models, rather than promoting harmful content.

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References

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. 2016. Layer normalization. *stat*, 1050:21.

Yonatan Belinkov and James Glass. 2019. Analysis methods in neural language processing: A survey. *Transactions of the Association for Computational Linguistics*, 7:49–72.

Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. 2023. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*.

Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, pages 610–623.

Christopher Bishop. 2006. Pattern recognition and machine learning. *Springer google schola*, 2:531–537.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac

Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. 2023. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*. Https://transformer-circuits.pub/2023/monosemantic-features/index.html.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Hila Chefer, Idan Schwartz, and Lior Wolf. 2022. Optimizing relevance maps of vision transformers improves robustness. *Advances in Neural Information Processing Systems*, 35:33618–33632.

Kevin Clark. 2017. Computing neural network gradients.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. 2023. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*.

Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2022. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8493–8502.

Guy Dar, Mor Geva, Ankit Gupta, and Jonathan Berant. 2022. Analyzing transformers in embedding space. *arXiv preprint arXiv:2209.02535*.

Alexander Yom Din, Taelin Karidi, Leshem Choshen, and Mor Geva. 2023. Jump to conclusions: Short-cutting transformers with linear transformations. *arXiv preprint arXiv:2303.09435*.

Yogesh K Dwivedi, Nir Kshetri, Laurie Hughes, Emma Louise Slade, Anand Jeyaraj, Arpan Kumar Kar, Abdullah M Baabdullah, Alex Koohang, Vishnupriya Raghavan, Manju Ahuja, et al. 2023. "so what if chatgpt wrote it?" multidisciplinary perspectives on opportunities, challenges and implications of generative conversational ai for research, practice and policy. *International Journal of Information Management*, 71:102642.

N Elhage, N Nanda, C Olsson, T Henighan, N Joseph, B Mann, A Askell, Y Bai, A Chen, T Conerly, et al. 2021. A mathematical framework for transformer circuits.

Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. 2022. Toy models of superposition.

Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. 2023. Dissecting recall of factual associations in auto-regressive language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12216–12235, Singapore. Association for Computational Linguistics.

Mor Geva, Avi Caciularu, Guy Dar, Paul Roit, Shoval Sadde, Micah Shlain, Bar Tamir, and Yoav Goldberg. 2022a. LM-debugger: An interactive tool for inspection and intervention in transformer-based language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 12–21, Abu Dhabi, UAE. Association for Computational Linguistics.

Mor Geva, Avi Caciularu, Kevin Wang, and Yoav Goldberg. 2022b. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 30–45, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2021. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495.

Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. 2024. Patchscope: A unifying framework for inspecting hidden representations of language models. *arXiv preprint arXiv:2401.06102*.

Almog Gueta, Elad Venezian, Colin Raffel, Noam Slonim, Yoav Katz, and Leshem Choshen. 2023. Knowledge is a region in weight space for fine-tuned language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1350–1370, Singapore. Association for Computational Linguistics.

Adi Haviv, Ido Cohen, Jacob Gidron, Roei Schuster, Yoav Goldberg, and Mor Geva. 2023. Understanding transformer memorization recall through idioms. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics, EACL 2023, Dubrovnik, Croatia, May 2-6, 2023*, pages 248–264. Association for Computational Linguistics.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. 2022. Editing models with task arithmetic. In *The Eleventh International Conference on Learning Representations*.

Sakshi Indolia, Anil Kumar Goswami, and Pooja Asopa. 2018. Conceptual understanding of convolutional neural network-a deep learning approach. *Procedia computer science*, 132:679–688.

Shahar Katz and Yonatan Belinkov. 2023. VISIT: Visualizing and interpreting the semantic information flow of transformers. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 14094–14113, Singapore. Association for Computational Linguistics.

Bobak Kiani, Randall Balestriero, Yann LeCun, and Seth Lloyd. 2022. projunn: efficient method for training deep networks with unitary matrices. *Advances in Neural Information Processing Systems*, 35:14448–14463.

DP Kingma. 2014. Adam: a method for stochastic optimization. In *Int Conf Learn Represent*.

Y Le Cun. 1988. A theoretical framework for backpropagation. In *Proceedings of the 1988 Connectionist Models Summer School*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 36.

Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2023. Mass-editing memory in a transformer. *International Conference on Learning Representations*.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2016. Pointer sentinel mixture models. In *International Conference on Learning Representations*.

Vivek Miglani, Aobo Yang, Aram Markosyan, Diego Garcia-Olano, and Narine Kokhlikyan. 2023. Using captum to explain generative language models. In *Proceedings of the 3rd Workshop for Natural Language Processing Open Source Software (NLP-OSS 2023)*, pages 165–173.

Beren Millidge and Sid Black. 2022. The singular value decompositions of transformer weight matrices are highly interpretable.

Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D Manning. 2021. Fast model editing at scale. In *International Conference on Learning Representations*.

nostalgebraist. 2020. interpreting gpt: the logit lens.

Chris Olah, Nick Cammarata, Chelsea Voss, Ludwig Schubert, and Gabriel Goh. 2020. Naturally occurring equivariance in neural networks. *Distill*, 5(12):e00024–004.

Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. 2018. Improving language understanding by generative pre-training.

Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners. OpenAI blog.

Ori Ram, Liat Bezalel, Adi Zicher, Yonatan Belinkov, Jonathan Berant, and Amir Globerson. 2023. What are you token about? dense retrieval as distributions over the vocabulary. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2481–2498, Toronto, Canada. Association for Computational Linguistics.

David E Rumelhart, Geoffrey E Hinton, and Ronald J Williams. 1986. Learning representations by back-propagating errors. *nature*, 323(6088):533–536.

Wojciech Samek, Thomas Wiegand, and Klaus-Robert Müller. 2017. Explainable artificial intelligence: Understanding, visualizing and interpreting deep learning models. *CoRR*, abs/1708.08296.

Soumya Sanyal and Xiang Ren. 2021. Discretized integrated gradients for explaining language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 10285–10299.

Gabriele Sarti, Nils Feldhus, Ludwig Sickert, and Oskar van der Wal. 2023. Inseq: An interpretability toolkit for sequence generation models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 421–435, Toronto, Canada. Association for Computational Linguistics.

Adi Simhi and Shaul Markovitch. 2023. Interpreting embedding spaces by conceptualization. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1704–1719, Singapore. Association for Computational Linguistics.

K Simonyan, A Vedaldi, and A Zisserman. 2014. Deep inside convolutional networks: visualising image classification models and saliency maps. In *Proceedings of the International Conference on Learning Representations (ICLR)*. ICLR.

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, et al. 2023. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. Axiomatic attribution for deep networks. In *International conference on machine learning*, pages 3319–3328. PMLR.

Alex Tamkin, Mohammad Taufeeque, and Noah D Goodman. 2023. Codebook features: Sparse and discrete interpretability for neural networks. *arXiv preprint arXiv:2310.17230*.